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Radiation source assembly and apparatus for layer-by-layer formation of three-dimensional objects

Abstract

A radiation source assembly (**100**) for an apparatus (**110**) for layer-by-layer formation of a three-dimensional object by the consolidation of particulate material, the radiation source assembly (**100**) comprising: a plurality of radiation sources (**21**), each radiation source (**21**) being operable to emit a respective beam of radiation (**26**) towards a build bed surface (**12**) of the apparatus (**110**); and one or more collimators (**22**) arranged to collimate the beams of radiation (**26**) to produce one or more collimated beams of radiation (**27**) and to direct said collimated beams (**27**) of radiation towards the particulate material on the build bed surface (**12**).

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

(1) This application is the U.S. national phase of PCT Application No. PCT/GB2020/05167, filed on Jun. 5, 2020, which claims priority to GB 1908186.8 filed Jun. 7, 2019, the disclosures of which are herein incorporated by reference in their entirety.

FIELD OF THE INVENTION

(2) The present disclosure relates to a radiation source assembly for an apparatus for the layer-by-layer formation of three-dimensional (3D) objects from particulate material, and to the operation of such a radiation source assembly within the apparatus. The radiation source assembly may be particularly suitable for particulate material bed applications that require reliable control of the temperature and location of the heating of the particulate material within the apparatus.

BACKGROUND

(3) Processes used to make three-dimensional objects from particulate material, such as Laser Sintering or High Speed Sintering, are receiving increased interest as they move towards faster throughput times and become industrially viable. In these processes, the object is formed layer-by-layer from successive layers of particulate material that are distributed across a work surface. The work surface comprises a build bed surface and such processes involve applying heat to successively precondition (or pre-heat) and consolidate (fuse, melt or sinter) particulate material that is distributed to form a layer on the build bed surface. Each layer of particulate material is consolidated over defined regions of the build bed surface in accordance with image data in order to form a ‘slice’ of the three-dimensional object. The build bed is then lowered, a new layer of particulate material is distributed across the work surface, and the process is repeated.

(4) Ensuring reliable and controllable heating of the distributed layers plays an important role in determining the quality of the finished object. For example, preferably there should be no variations in heating of the particulate layer that affect the integrity, or undesirably vary the properties, of the ‘slice’ that is being formed. Control of the temperature of the particulate material surface is important for a variety of reasons, such as ensuring controlled temperature conditions across the entire build bed surface, and from layer to layer, so as to avoid undesirable consolidation conditions, such as “curl” where consolidated material moves out of the plane of the build surface as the build cake cools down. It is therefore necessary to control the intensity and duration of the heating of the particulate material, whilst ensuring that the amount of heat provided to the layer is sufficient to consolidate the particulate material in the desired areas.

(5) Laser Sintering, which may use polymeric or metal powders, uses a laser to trace the shape of a slice of the three-dimensional object in the particulate material, sintering the particulate layer. Another layer of particulate material is then deposited and the shape of the next slice of the object is traced by the laser, and so on, to fabricate a three-dimensional object. A similar process using an electron beam may be used to fuse particulate material such as metal powders (Electron Beam Sintering).

(6) In contrast to Laser Sintering or Electron Beam Sintering, where the energy source is required to trace the shape of the object in each layer of particulate material, a High Speed Sintering process may be used. In this process a radiation absorbing material (RAM) is deposited in the shape of each slice of the three-dimensional object onto the layer of particulate material, in one or more passes of a droplet deposition head or array of droplet deposition heads (such as printheads). Then, each layer is irradiated with a radiation source, for example an infrared light, across the entire build bed surface, such that only particulate material to which the RAM has been applied is consolidated to form the slice.

(7) Currently in High Speed Sintering processes a combination of fixed heaters (generally overhead heaters such as ceramic heaters) above the build bed surface and lamps mounted on a movable sled system (scanning lamps) are widely used. The scanning lamps are generally infra-red (IR) emitting and are switched ON prior to the pass and remain ON during the entire pass over the build bed

surface. This scanning pass may be done to pre-heat the particulate material and/or to consolidate it. Such an arrangement is undesirable for a number of reasons: firstly because it uses a lot of power (lamp power between 1 kW to 3 kW for consolidating and for pre-heating), and secondly because the entire build bed surface is heated to a near consolidation temperature. This can cause the particulate material that has not had RAM deposited on it to become harder, and consequently "de-caking" (removal of the excess particulate material from the formed part) can become difficult. A hard "cake" can make post processing of the formed parts arduous, by requiring significant manual labour. This in turn leads to increased part cost, e.g. due to longer turnaround time/higher labour time. A further undesirable side-effect of the generalised heating is that the unconsolidated particulate material degrades at a faster rate than if it had not been heated to such high temperatures. The reduction in particulate material quality reduces its suitability for re-use (unused particulate material is generally recycled in the apparatus) and can thus make the 3D printing process more expensive.

(8) It is an object of the present invention to improve processes for the layer-by-layer formation of three-dimensional (3D) objects from particulate material, such as the High Speed Sintering process, to ensure more reliable and consistent control of the pre-heating and consolidation steps so as to overcome the above described disadvantages.

SUMMARY

(9) Aspects of the invention are set out in the appended independent claims, while details of particular embodiments of the invention are set out in the appended dependent claims.

(10) According to a first aspect of the disclosure there is provided a radiation source assembly for an apparatus for the layer-by-layer formation of a three-dimensional object by the consolidation of particulate material, the radiation source assembly comprising: a plurality of radiation sources, each radiation source being operable to emit a respective beam of radiation towards a build bed surface of the apparatus; and one or more collimators arranged to collimate the beams of radiation to produce one or more collimated beams of radiation and to direct said collimated beams of radiation towards the particulate material on the build bed surface.

(11) According to certain embodiments the radiation source assembly further comprises one or more focusing lenses arranged to focus the collimated beams of radiation onto the particulate material on the build bed surface, wherein said one or more collimators are arranged between the radiation sources and the focusing lenses.

(12) According to certain embodiments, one or more radiation source assemblies according to the first aspect of the disclosure may be controllable so as to provide targeted pre-heating and/or consolidation of particulate material on the build bed surface.

(13) According to certain embodiments, one or more radiation source assemblies according to the first aspect of the disclosure may be controllable so as to alter the focus, spot size, intensity, duty cycle or wavelength of the beams of radiation.

(14) According to a second aspect of the disclosure there is provided an apparatus for the layer-by-layer formation of a three-dimensional object by the consolidation of particulate material, the apparatus comprising one or more radiation source assemblies according to the first aspect of the disclosure, for pre-heating and/or consolidating the particulate material on the build bed surface.

(15) According to a third aspect of the disclosure there is provided a method for manufacturing a three-dimensional object from a particulate material using a radiation source assembly according to the first aspect of the disclosure.

(16) According to a fourth aspect of the disclosure there is provided a radiation system controller operable to cause the radiation source assembly according to the first aspect of the disclosure to carry out the method according to the third aspect of the disclosure.

(17) The present disclosure also provides a fixed pre-heater or pre-heater array above the build bed surface for use with any of the above aspects of the disclosure to provide targeted and controllable pre-heating of selected portions of the particulate material on the build bed surface.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Embodiments will now be described with reference to the accompanying Figures of which:

(2) FIG. **1***a* is a schematic diagram of a perspective view of a radiation source assembly for an apparatus for manufacturing a three-dimensional object, comprising one or more collimators;

(3) FIG. **1***b* is a schematic diagram of a perspective view of another radiation source assembly for an apparatus for manufacturing a three-dimensional object, comprising a focusing lens array and one or more collimators;

(4) FIG. **1***c* is a schematic diagram of a perspective view of another radiation source assembly for an apparatus for manufacturing a three-dimensional object, comprising one or more focusing lenses;

(5) FIG. **2***a* is a schematic diagram of a cross-section through an apparatus for the layer-by-layer formation of three-dimensional objects by the consolidation of particulate material;

(6) FIG. **2***b* is a schematic diagram of a cross-section through a portion of the apparatus depicted in FIG. **2***a* incorporating a radiation source assembly similar to that depicted in FIG. **1***b;*

(7) FIG. **3***a* is a schematic top view of two scanning sleds suitable for use in the apparatus depicted in FIGS. **2***a* and **2***b;*

(8) FIG. **3***b*, depicts a process for deposition and consolidation utilising the scanning sled layout illustrated in FIG. **3***a;*

(9) FIG. **3***c* is a schematic diagram of the printing sled at four time-steps during the process depicted in FIG. **3***b;*

(10) FIG. **3***d* is a schematic diagram of the distribution sled at four time-steps during the process depicted in FIG. **3***b;*

(11) FIGS. **4***a* and **4***b* depict ways in which a radiation source (and optionally a pre-heater) for a test design for an apparatus for manufacturing a three-dimensional object can be operated in order to pre-heat or consolidate particulate material to form a slice of a three-dimensional object;

(12) FIGS. **5***a* and **5***b* depict ways in which a radiation source assembly can be operated in a targeted manner in order to pre-heat or consolidate particulate material to form a slice of a three-dimensional object;

(13) FIG. **6** depicts ways in which a radiation source assembly can be divided into two individually addressable groups and operated in a targeted manner in order to pre-heat or consolidate particulate material to form a slice of a three-dimensional object; and

(14) FIG. **7** depicts a way in which an overhead pre-heater can be operated in a targeted manner in order to pre-heat discrete regions of particulate material on the build bed surface.

(15) It should be noted that the drawings are not to scale and that certain features may be shown with exaggerated sizes so that these are more clearly visible.

DETAILED DESCRIPTION

(16) Embodiments and their various implementations will now be described with reference to the drawings. Throughout the following description, like reference numerals are used for like elements where appropriate.

(17) FIG. **1***a* is a schematic diagram of a perspective view of a radiation source assembly **100** for an apparatus **110** for manufacturing a three-dimensional object, comprising one or more collimators **22**. More particularly, FIG. **1***a* shows a radiation source assembly **100** for an apparatus **110** for manufacturing a three-dimensional object by the consolidation of particulate material. Further details of the apparatus **110** are depicted in FIG. **2**. As can be seen in FIG. **1***a*, the radiation source assembly **100** comprises a plurality of radiation sources **21**(*i*-*vi*), each radiation source **21** being operable to emit a respective beam of radiation **26** towards a build bed surface **12** of the apparatus **110** for pre-heating and/or consolidating particulate material; and a plurality of collimators **22**(*i*-*vi*)

arranged between the plurality of radiation sources **21**(*i*-*vi*) and the build bed surface **12** so as to collimate the beams of radiation **26** to produce collimated beams of radiation **27** and to direct said collimated beams of radiation **27** towards the particulate material on the build bed surface **12**. The use of collimator(s) may be advantageous because collimating the beams **26** emitted by the plurality of radiation sources **21**(*i*-*vi*) may reduce or eliminate beam diversion and reduce the likelihood of overlap of beams from different radiation source(s) **21**. Such overlap is undesirable because it means that some regions of the build bed surface **12** may receive more radiation energy than others, leading to ‘hot’ and ‘cold’ spots and potentially causing uneven formation of a three dimensional object.

(18) It is further advantageous to collimate the beams **26** as this arrangement provides greater flexibility in positioning the radiation source assembly **100** such that it can be moved further from the build bed surface **12** and reduces the likelihood of undesirable heating of the radiation source assembly **100** by heat reflected from the build bed surface **12**. Additionally, moving the radiation source assembly **100** further away from the build bed surface **12** may reduce the amount of loose particulate material that settles on (and possibly adheres to) said assembly, thereby extending its life or reducing the number of cleaning operations required.

(19) Moreover, in order to fine tune the design and operation of the radiation source assembly **100** it may be desirable to be able to alter the position of the plurality of radiation sources **21**(*i*-*vi*). As can be seen in FIG. **1***a* the radiation source assembly **100** is located above the build bed surface **12** in the Z-direction and arranged so as to span the build bed **16** in the Y-direction. The plurality of radiation sources **21**(*i*-*vi*) are located above the build bed surface **12** at a distance h**1** in the Z-direction and mounted to a radiation source array holder **20**. In some implementations, the distance h**1** of the radiation sources **21**(*i*-*vi*) from the build bed surface **12** may be adjustable. The collimators **22**(*i*-*vi*) are located above the build bed surface **12** at a distance h**2** in the Z-direction and may be mounted in any suitable manner so as to collimate the beams of radiation **26** such that the collimated beams **27** impinge upon the build bed surface **12**.

(20) Further it may be desirable to be able to alter the position above the build bed surface **12** of the one or more collimators **22**, either in tandem with, or independently of the plurality of radiation sources **21**(*i*-*vi*), in order to fine tune the design and operation of the radiation source assembly **100**. The distance h**2** of the one or more collimators **22** from the build bed surface **12** may be adjustable. In some embodiments, it may be desirable to be able to individually alter the position of each individual collimator **22**(*i*-*vi*).

(21) The radiation source assembly **100** may comprise one or more collimators **22**. FIG. **1***a* depicts a plurality of collimators **22**(*i*-*vi*) which comprise collimating lenses. The one or more collimating lenses may be spherical lenses. There are also present a plurality of radiation sources **21**(*i*-*vi*) such that there is a respective collimator **22** for each of the plurality of radiation sources **21**(*i*-*vi*), but this 1:1 relationship is not essential and other implementations may be used. An arrangement with collimators and radiation sources as per FIG. **1***a* may be suitable for pre-heating, but in some embodiments it may also be suitable for consolidating particulate material on the build bed surface **12**. In some implementations, there may be one collimator **22** for all the radiation sources **21**, and in still further implementations the plurality of radiation sources **21**(*i*-*vi*) may be arranged into a plurality of individually addressable groups and there may be a respective collimator **22** for each of the plurality of groups of radiation sources **21**, or any other ratio or combination of collimators **22** and radiation sources **21** as may be suitable.

(22) As FIG. **1***a* shows, the radiation source assembly **100** comprises a plurality of radiation sources **21**(*i*-*vi*) which, in this instance, are arranged in a radiation source array **19** as a single row. In other implementations, the radiation source array **19** may comprise a plurality of rows of radiation sources **21** so as to form a multi-row array, or grid. Such an arrangement may enable more of the build bed surface **12** to be addressed at once, which may be beneficial for uniform heating and for speed of object formation. In some implementations, the radiation source assembly

100 may comprise a radiation source array **19** where the rows of radiation sources **21** are arranged in a staggered manner. Such a staggered arrangement may be beneficial to ensure thorough and consistent heating of the build bed surface **12**. Any suitable or practicable stagger pattern may be used.

(23) FIG. **1***a* additionally depicts a radiation system controller **53** which controls the radiation source assembly **100** in accordance with image data for the three-dimensional object so as to switch the radiation sources **21** ON or OFF as they are moved over the build bed surface **12**, such that individual radiation sources are individually addressable and/or controllable. Alternatively, the plurality of radiation sources **21**(*i*-*vi*) may be arranged into a plurality of individually addressable groups and sub-groups, such that the radiation system controller **53** provides signals to control the individual groups and/or sub-groups independently from one another. For example, individual radiation sources **21** or groups/sub-groups of radiation sources **21** may be switched ON/OFF depending on image data and/or manufacturing data supplied to the radiation system controller **53** that indicates the locations on the build bed surface **12** which require a pre-heat step or a consolidation step. Further, the individual radiation sources **21** or groups/sub-groups of radiation sources **21** may be operated to produce different intensities of radiation at the build bed surface **12** depending on whether the radiation source assembly **100** is being used for a pre-heating or consolidation step. Still further, the intensity of radiation emitted by the radiation source assembly **100** or individual radiation sources **21** or groups/sub-groups of radiation sources **21** may optionally be controlled in response to measurements of the temperature of the build bed surface **12**, for example so that hotter regions of the build bed surface are pre-heated or consolidated as desired without being over-heated. Such measurements could be made, for instance, by an optional temperature sensor **205** such as a thermal camera, or pyrometer, located above the build bed surface **12**, or by some other suitable measurement device or method.

(24) In some implementations, in addition to/instead of altering the intensity of the radiation being emitted by the radiation sources **21**, the wavelength of the radiation may be altered depending on whether a pre-heating or consolidation step is being performed, or in response to measurements of the temperature of the build bed surface **12**. This may be achieved by, for example, having groups or sub-groups comprising different types of radiation sources **21** within the radiation source assembly **100**, for example different types of light emitting diodes (LEDs) that emit different wavelengths, laser diodes, monochromatic lamps. Another suitable radiation source **21** could be one or more infrared sources coupled with one or more digital mirror devices (DMDs).

(25) Turning now to FIG. **1***b*, this is a schematic diagram of a perspective view of another radiation source assembly **101** for an apparatus for manufacturing a three-dimensional object. FIG. **1***b* comprises many features depicted in the apparatus of FIG. **1***a*, and like reference numerals have been used for like elements where appropriate. As can be seen, such a radiation source assembly **101** comprises one or more collimators **22**(*i*-*vi*) and further comprises a common (i.e. shared) focusing lens **23**. The one or more collimators **22**(*i*-*vi*) are arranged between the radiation sources **21**(*i*-*vi*) and the focusing lens **23** so as to collimate the beams of radiation **26** to produce a plurality of collimated beams **27**. The focusing lens **23** is arranged to focus the collimated beams **27** into focused beams **28** which are arranged to be focused onto particulate material on the build bed surface **12**.

(26) In some implementations, the focusing lens **23** may instead be one or more focusing lenses **23** which may be arranged in an array, for example as a row or rows to form an array of focusing lenses **23**. Further, in some implementations the focal length(s) of the one or more focusing lenses **23** may be adjustable and/or the spot size(s) may be adjustable.

(27) For example, a method of operating the radiation source assembly **101** may involve controlling the one or more focusing lenses **23** to adjust their focal length in order to focus and de-focus the beams **28**, and/or to alter the spot size. This method may be used to alter the amount of heat supplied to an area of the build bed surface **12**. This might be desirable in order to switch

between temperatures suitable for pre-heating and those suitable for consolidation. It may also be desirable to be able to fine-tune the energy supplied to the particulate material in response to temperature measurements, for example, by altering how tightly the beams **28** are focused and hence the spot size and therefore energy supplied per unit area to particulate material on the build bed surface **12**.

(28) The one or more focusing lenses **23** are located at a distance h**3** from the build bed surface **12**, which in some implementations may be adjustable. In some implementations, it may be desirable to alter the distances h**1**, h**2** and h**3** independently of each other such that the distance h**1** of the radiation sources **21** from the build bed surface **12** and the distance h**2** of the one or more collimators **22** and the distance h**3** of the one or more focusing lenses **23** from the build bed surface **12** may be independently adjustable.

(29) Since the plurality of radiation sources **21**(*i-vi*) may be individually addressable and controllable, in some implementations the intensity of the radiation being emitted and/or the wavelength of the radiation being emitted by individual radiation sources **21** and/or the duty cycle of the plurality of radiation sources **21**(*i-vi*) may be controlled and a method of operation of the radiation source assembly **101** may involve controlling these features/parameters.

(30) In some implementations, the focal length and/or the spot size on the build bed surface may be altered and controlled by adjusting the one or more focusing lenses **23** in any manner that is suitable to achieve the desired temperature at the build bed surface **12** for the pre-heating or consolidation of a three-dimensional object. For example the duty cycle may be controlled to alter the length of time that the radiation sources **21** are switched ON in response to overall temperature changes in the apparatus **110**, or to select which groups or sub-groups of radiation sources **21** are switched ON. Further, in some implementations, the focal length(s) of the one or more focusing lenses **23** may be individually adjusted to control the intensity of the focused beams **28** on the build bed surface **12**, and/or to alter the spot size. The one or more focusing lenses **23** may also be arranged into one or more groups or sub-groups that may correspond to one or more of the groups or sub-groups of radiation sources **21** such that the controller **53** can control the focal length or spot size of a group or sub-group of radiation sources. FIG. **1***b* shows one focusing lens **23** to a row of radiation sources **21**(*i-vi*) but it may be understood that this is in no way limiting and in other implementations any suitable configuration of focusing lenses and radiation sources may be used.

(31) A method of operation of the radiation source assembly **101** depicted in FIG. **1***b* may therefore involve arranging the plurality of radiation sources **21**(*i-vi*) into a plurality of individually addressable groups, each group comprising one or more of the plurality of radiation sources **21**(*i-vi*). The method may therefore involve controlling the intensity of the radiation being emitted and/or the wavelength of the radiation being emitted and/or the spot size and/or the duty cycle of the group of radiation sources **21**. For example, the radiation source assembly **100**/**101** may comprise different types of LEDs, operable to emit different wavelengths of radiation, and the method may involve arranging each type of LED into a separately controllable group or sub-group and choosing which group/sub-group to switch

(32) ON depending on whether pre-heating or consolidation steps are being performed. Further, the method of operation may involve altering the focal length of the one or more focusing lenses **23** so as to alter the intensity of the focused beams **28** at the build bed surface **12**, or to alter the spot size at the build bed surface **12**.

(33) FIG. **1***c* is a schematic diagram of a perspective view of a radiation source assembly **102** for an apparatus **110** for manufacturing a three-dimensional object according to a variant comprising a common (i.e. shared) focusing lens **23**, as in FIG. **1***b*, but without the one or more collimators. FIG. **1***c* comprises many features depicted in the embodiments of FIGS. **1***a* and **1***b*, and like reference numerals have been used for like elements where appropriate. More particularly, FIG. **1***c* shows a radiation source assembly **102** for an apparatus **110** for manufacturing a three-dimensional object by the consolidation of particulate material. As can be seen in FIG. **1***c*, the radiation source

assembly **102** comprises a plurality of radiation sources **21**(*i-vi*) and a common focusing lens **23** arranged to focus a plurality of beams of radiation **26** into focused beams **28** which are focused onto particulate material on the build bed surface **12**. In other implementations, the focusing lens **23** may instead be one or more focusing lenses **23** which may be arranged in a row or rows to form an array of focusing lenses **23**.

(34) Turning now to FIGS. **2***a* and **2***b*, FIG. **2***a* is a schematic diagram of a cross-section through an apparatus **110** for manufacturing a three-dimensional object from a particulate material, comprising one or more radiation source assemblies **100**/**101**/**102** as described in FIGS. **1***a-c*, that are mounted to a sled system **30**, for pre-heating and/or consolidating particulate material on the build bed surface **12**. The apparatus **110** further comprises a work surface **13**, which comprises the build bed surface **12** and a working space **4** in which the sled system **30**, comprising sleds **32** and **34**, an (optional) overhead fixed heater **29**, and a build bed **16** are arranged. The build bed surface **12** is contiguous with the work surface **13** and forms the uppermost level of the build bed **16** in the Z-direction. The work surface **13** is the surface onto which particulate material is dosed from the particulate material dosing system **40**. The apparatus **110** therefore further comprises a particulate material dosing system **40** which comprises a dosing device **416** for depositing particulate material onto the work surface **13**. The dosing device **416** may be a dosing blade, for example.

(35) The build bed **16** and the formed part of the three-dimensional object **2** are supported inside a build chamber **10**, in which the build bed **16** is supported by a piston **18** which is movable in the Z-direction. The piston **18** lowers the build bed **16** by a layer height each time a slice of the three-dimensional object **2** is completed. Moveable across the build bed surface **12** is a sled system **30**, supporting particulate material distribution and processing equipment, in this implementation comprised within two sleds **32** and **34**. One sled is a printing sled **32** supporting one or more droplet deposition heads and a radiation source assembly (not shown) such as those depicted in FIGS. **1***a-c*. The other sled is a distribution sled **34** comprising a particulate material distribution device **37**, for example a roller or a blade. Each sled is moveable on a rail **36** across the build bed surface **12**. Particulate material that has been dosed to the work surface **13** can be distributed across the build bed surface **12** by the distribution device **37** mounted on the distribution sled **34**. Excess particulate material may be pushed into the return slot **426** by the distribution device **37** and returned to the dosing system **40** via the return tube **427**. Once a layer of particulate material has been distributed over the build bed surface **12** by the distribution device **37**, the printing sled **32** moves across the layer to allow the one or more droplet deposition heads to deposit radiation absorbing material (RAM), in accordance with image data. Next, the radiation source assembly (not shown) mounted on the printing sled **32** is used to irradiate the particulate material so as to consolidate the areas onto which RAM has been deposited. Next, the piston **18** lowers the build bed **16** by a layer thickness, and the process is repeated until the three-dimensional object **2** is fully built.

(36) FIG. **2***b* is a schematic diagram of a cross-section through a portion of an apparatus **110** incorporating a radiation source assembly **101** similar to that depicted in FIG. **1***b*. The radiation source assembly **101** is mounted to the printing sled **32**. The printing sled **32** may further comprise a droplet deposition head array **35** comprising one or more droplet deposition heads so as to span the build bed **16** in the Y-direction (into the page). The apparatus **110** for manufacturing a three-dimensional object accordingly comprises one or more droplet deposition heads which are operable to deposit a pattern of radiation-absorbing material (RAM) onto a layer of particulate material **71** on the build bed surface **12** in order to define the cross-section of the three dimensional object in said layer.

(37) The droplet deposition head array **35** may be controlled by a printer controller **51** to deposit RAM in accordance with image data that may be supplied from an overall process controller **60** or that may be encoded in the printer controller **51**. In operation, the printing sled **32** moves across the build bed surface **12** and the droplet deposition heads deposit a radiation absorbing material (RAM)

so as to describe the shape of the relevant slice of the geometry of the three-dimensional object(s) to be formed, in accordance with image data. The layer of particulate material **71** is then irradiated using some or all of the radiation sources **21** in the radiation source array **19** in order to consolidate the regions of particulate material **71** that have had RAM deposited onto them. The radiation source array **19** may be controlled by a radiation system controller **53** which, in accordance with image data, enables individual radiation sources **21** to be activated in a controllable manner so as to switch them ON at the appropriate location, for an appropriate length of time and at an appropriate wavelength, focal length, spot size and intensity in order to achieve the desired consolidation effect. The radiation source assembly may be arranged so that the plurality of radiation sources are arranged into a plurality of individually addressable groups and sub-groups thereof with the radiation system controller **53** controlling the groups and sub-groups. Alternatively the entire build bed surface **12** may be irradiated if desired. In some implementations, it is preferable to irradiate substantially only the areas where RAM has been deposited, such that only the particulate material to which the RAM has been applied is consolidated. It may be understood that for some designs, and to prevent undesirable effects such as edge curl in the three-dimensional object being created, particulate material outside of the region on which the RAM was deposited may also be irradiated in a controlled manner so as to control the temperature gradient at the edge of the object, to ensure the object does not cool and solidify in situ and cause curl that could lead to catastrophic build failure. It may further be understood that a pre-heat step may be used to pre-heat selected portions of the build bed surface **12** and that this step may be performed by the radiation source assembly **100**/**101**/**102**. The radiation source assembly **100**/**101**/**102** may be controlled by the radiation system controller **53**.

(38) Considering FIG. **2***b* further, a distribution sled **34** is located ahead of the printing sled **32** when viewed along the X-direction. The distribution sled **34** has a particulate material distribution device **37**. In operation, the particulate material distribution device **37** acts to distribute the particulate material **70** evenly across the build bed surface **12** into a layer of particulate material **71** with controllable thickness. The particulate material distribution device **37** may be one or more of a roller and/or a blade or a combination of roller(s) and/or blade(s) or any other arrangement of devices suitable to distribute the particulate material across the build bed surface **12**. The thickness of the layer of particulate material **71** will depend on the particulate material being used and on the distance by which the build bed **16** is being lowered ahead of distributing the particulate material. However, a typical layer thickness in a High Speed Sintering Machine may be around 50 μm-2mm; in some implementations, the layer thickness may be 80-100 μm.

(39) The sleds **32**, **34** may be controlled by a sled system controller **52** so as to control the speed and direction of travel of the sleds, and the timing of the movement of the sleds **32**, **34** relative to each other.

(40) FIG. **2***b* further shows that the radiation system controller **53** may be controlled by an overall process controller **60** that may also control the sled system controller **52** and the printer controller **51** so that all of the various process steps may be controlled and co-ordinated. The system may further comprise one or more temperature sensors **205**. Such sensors **205** may, for example, be thermal cameras or pyrometers that monitor the temperature of the build bed surface **12** and provide feedback to the controller **60** to enable the process steps and the operation of the radiation source assembly **100**/**101**/**102**, the sled system **30** and the droplet deposition head array **35** to be controlled and optimised. In this way, the desired temperature profile across the particulate material layer **71** may be maintained.

(41) In FIG. **2***b*, the sensor **205** is located on the printing sled **32**, but other suitable locations may be used, such as amongst the radiation sources **21**, or at some suitable fixed location, such as above the build bed surface **12**, or in any other suitable location where they are able to record the necessary data. The sensors **205** may optionally comprise an array of sensors **205** distributed suitably, for example at a number of locations above the build bed surface **12**, so as to monitor the

temperature of the smoothed particulate material **71** over some or all of the build bed surface **12**. Alternatively an array of sensors **205** may be distributed over the printing sled **32**, for example at the front and rear of the printing sled **32**, so as to monitor the temperature of the particulate material layer **71** before and after irradiation. There may also be sensors **205** located before and after the printhead array **30**, in the X-direction, so as to monitor any temperature changes due to liquid deposition in any printing step.

(42) Attention is now drawn to FIG. **3***a*, which is a schematic top view of a sled system **30** which may be used as part of the apparatus **110**. In this implementation, the sled system **30** comprises a distribution sled **34** and a printing sled **32** arranged on bearings **38** on rails **36**, with the printing sled **32** located ahead of the distribution sled **34** in the X-direction. The rails **36** act to suspend the sleds **32**, **34** above a work surface **13** of the apparatus **110**. In this implementation, the droplet deposition head array **35** comprises three droplet deposition heads **33** spanning the width of the build bed surface **12**, in the Y-direction, in a staggered configuration, though this is by no means limiting and the number of droplet deposition heads **33** may be adjusted as necessary so as to span the width of any specific build bed surface **12** in the Y-direction. To ensure sufficient laydown of RAM there may be more than one row of droplet deposition heads **33** in the X-direction. The apparatus **110** may further comprise one or more pre-heaters **24** for pre-heating particulate material on the build bed surface **12** so that the one or more radiation source assemblies **101** is/are operable to consolidate particulate material on the build bed surface **12**. It should be understood that whilst FIG. **3***a* depicts radiation source assembly **101**, this is not limiting, and in other implementations radiation source assemblies **100** or **102** could be used as well/instead.

(43) In the embodiment, depicted in FIG. **3***a* the pre-heater **24** is located on the distribution sled **34**, though this is by no means limiting. Such a pre-heater **24** may be a halogen lamp, or a ceramic lamp, or a further radiation source assembly **100**/**101**/**102** or any other suitable heat source. In further implementations, the pre-heater may be a separate heater or heaters **29** located above the build bed surface **12**, such as an overhead array of ceramic lamps, halogen lamps or LEDs. Still further, the pre-heater may comprise one or more sled-mounted pre-heaters **24** and one or more pre-heaters **29** located above the build bed surface **12**. In implementations, where there are one or more separate pre-heaters **24**/**29**, these may perform all of the pre-heat steps while the first one or more radiation source assemblies **100**/**101**/**102** may be used solely for consolidating particulate material on the build bed surface **12**. In other implementations, the pre-heat step(s) may be shared between the one or more pre-heaters **24**/**29** and the radiation source assembly **100**/**101**/**102** in any manner suitable for achieving the desired temperatures in the particulate material layer **71** on the build bed surface **12**.

(44) Turning now to FIG. **3***b*, this depicts a process for depositing RAM and consolidating particulate material using the sled layout as illustrated in FIG. **3***a*. The sleds **34**, **32** start at sled position **2** at step S**601**, with both the distribution sled **34** and the printing sled **32** arranged behind the dosing device **416**, such that the dosing device **416** is positioned between the sleds **34**, **32** and the build bed surface **12**. A first layer **71** of particulate material has already been distributed across the build bed surface **12** in a previous step.

(45) At step S**602**, the printing sled **32** is moved in the X-direction from the side of the dosing device **416** across the build bed surface **12** to the side of the return slot **426**. This is the printing sled **32** forward stroke. As the printing sled **32** moves across the build bed surface **12**, the droplet deposition heads **33** deposit droplets **31**, comprising RAM, in accordance with image data, onto the layer of particulate material **71**. The droplet deposition heads **33** may be printheads and the deposition process may be a printing process. The droplets **31** may comprise a fluid comprising RAM, or a carrier fluid and a material comprising RAM, or any other suitable combination of liquids and/or materials such that RAM is deposited on the build bed surface **12** in accordance with image data. In other implementations, other suitable apparatus may be used instead of the droplet deposition heads **33**, such that fluids and/or materials comprising RAM are delivered to the build

bed surface. For example, the RAM could be a particulate material, such as a powder, and may be deposited by a coater arranged above the build bed **16**.

(46) As the printing sled **32** moves across the build bed surface **12**, the radiation source assembly **101** which is mounted behind the droplet deposition heads **33** on the printing sled **32** (with respect to the X-direction of travel during step S**602**) consolidates the areas where RAM has been deposited. Once the printing sled **32** is clear of the dosing device **416** during step S**602**, or, if preferred, once the printing sled **32** has arrived at the opposite end of the work surface **13**, at step S**603**, the dosing device **416** is operated to dose a fresh pile of particulate material **70** up to the level of the work surface **13** along the full length of the dosing device **416**.

(47) Next, at step S**604** the distribution sled **34** is moved across the build bed surface **12** in the X-direction from the dosing device **416** side to the return slot **426** side—the distribution sled forward stroke. The distribution sled **34** passes over the dosing device **416**, and the distribution device **37** spreads the pile of particulate material **70** across the work surface **13**, distributing a layer **71** of particulate material on the build bed surface **12**, before pushing any excess particulate material **70** down the return slot **426**. The pre-heater **24** which is mounted on the distribution sled **34** behind the distribution device **37** (with respect to the direction of travel during step S**604**) optionally pre-heats the freshly distributed layer of particulate material **71**.

(48) At step S**605**, the floor of the build chamber **10** is lowered by the thickness of a layer of the object **2** using the piston **18**.

(49) At step S**606**, the distribution sled **34** is returned to the dosing device **416** side of the work surface **13**—the distribution sled **34** return stroke. After that, and before RAM is deposited onto the fresh layer of particulate material **71**, the floor of the build chamber **10** is raised. It may be raised to a level just short by the next layer thickness to be deposited with respect to the particulate material surface. The pre-heater **24** may optionally be utilised to help maintain the build bed surface **12** at a predetermined temperature.

(50) Finally, at step S**607**, the printing sled **32** is returned to the dosing device **416** side of the work surface **13**—the printing sled return stroke.

(51) Optionally, during the printing sled return stroke, the radiation source assembly **101** may be used as a pre-heating source to help maintain the build bed surface **12** at a predetermined temperature. (FIG. **3***a* depicts radiation source assembly **101**, but it should be understood that in other implementations radiation source assembly **100**/**102** may be used as well/instead.) The intensity and/or the wavelength of the radiation emitted by the radiation source assembly **100**/**101**/**102** may be adjustable for this function and/or the focal length of the one or more focusing lenses **23** (where present in the radiation source assembly) may be altered so as to alter the intensity of the focused beams **28** at the build bed surface **12** and/or to alter the spot size.

(52) The process steps described above form one of many possible processes for the layer-by-layer formation of three-dimensional objects. It may be understood that the locations of the various components on the various sleds in the sled system **30** can be altered and suitable orders of process steps devised as would be apt therefor. Process steps may be added, removed or reordered as necessary in order to achieve the desired result. The process controller **60** and/or the printer controller **51** and/or the sled system controller **52** and/or the radiation system controller **53** may be used in any suitable manner to control the chosen process steps.

(53) Turning now to FIG. **3***c*, this is a schematic diagram of the printing sled **32** at four time-steps t**1**-t**4** during the printing sled forward stroke S**602** as it passes over the build bed surface **12**. In FIG. **3***c*(*a*) at t**1**, the printing sled **32** comprising droplet deposition head array **35** has moved across the layer of particulate material **71** on the build bed surface **12** and the droplet deposition head array **35** has deposited radiation absorbing material (RAM) to form the next slice of the three dimensional object **2**. In this time-step, the radiation source assembly **100**/**101**/**102** has not yet passed over the region of deposited RAM on the particulate material layer **71**. In FIGS. **3***c*(*b*)-(*d*) the radiation source assembly **100**/**101**/**102** on the printing sled **32** moves over the build bed surface

12 so as to irradiate and consolidate the areas of particulate material layer **71** onto which RAM has been deposited to form a new slice of the three dimensional object **2**.

(54) It can be seen schematically from FIGS. **3***c*(*b*)-(*d*) that the individual radiation sources **21**, indicated by squares, may be operated in a controllable manner so as to be switched ON (dark square) when they are above the area where RAM has been deposited to define the three dimensional object **2**, and otherwise to be switched OFF (white squares). The length of time that each region of RAM on the build bed surface **12** is irradiated can therefore be carefully controlled, in accordance with the image data, so that all regions receive an appropriate amount of energy in order to be fully consolidated. This may prevent or reduce excessive heating of areas already consolidated, or reduce or eliminate heating of any regions where no RAM has been deposited.

(55) To this end, for example, the radiation system controller **53** may be used to control the radiation source assembly **100**/**101**/**102** so as to control the duty cycle (duration of irradiation) and/or the level of power supplied to the radiation sources **21**. The radiation sources **21** in the radiation source assembly **100**/**101**/**102** may be chosen so that they are operable to emit a plurality of wavelengths, and/or they are operable to emit radiation within an absorption band of wavelengths of the particulate material and/or of the radiation absorbing material (RAM) provided to the particulate material.

(56) In some implementations of the radiation source assembly **100**/**101**/**102**, the radiation source assembly **100**/**101**/**102** may comprise a first group of radiation sources **21** operable to emit radiation within an absorption band of wavelengths of the particulate material, and a second group of radiation sources **21** operable to emit radiation within an absorption band of wavelengths of radiation absorbing material (RAM) provided to the particulate material.

(57) In some implementations, the radiation source assembly **100**/**101**/**102** may comprise radiation sources **21** whereby the wavelength(s) of the emitted radiation is/are selected for pre-heating particulate material on the build bed surface **12**. Still further, in some implementations, the radiation source assembly **100**/**101**/**102** may comprise radiation sources **21** whereby the wavelength(s) of the emitted radiation is/are selected for consolidating particulate material on the build bed surface **12**.

(58) In some implementations, the wavelength(s) of the emitted radiation may be selected for both pre-heating and consolidating particulate material on the build bed surface **12**. For example, the radiation source assembly **100**/**101**/**102** may comprise two types of LEDs that emit light at different ranges of wavelengths, one range suitable for pre-heating and one suitable for consolidation, where the two types of LEDs are arranged into first and second groups. The two groups may be further divided into a plurality of sub-groups, for example based on their location, such that they can be addressed and controlled. For example individual sub-groups may be turned ON/OFF in accordance with image and processing data. Alternatively a type of radiation source, such as an LED, that emits radiation over a broad range of wavelengths, suitable for both pre-heating and consolidation, may be chosen.

(59) The particulate material may for example comprise powder, for example it may comprise polyamide (such as PA11, PA12, and PA6), polypropylene, polyurethane, other polymers, metals or ceramics. A suitable consolidation temperature for polyamide 12 (PA12), a commonly used powder in High Speed Sintering processes, may be >185° C., whereas a suitable consolidation temperature for polyamide 11 (PA11) may be >195° C. or possibly >200° C. Radiation suitable for consolidation may be in the IR or near IR spectrum and may have a wavelength of around 1 μm. The appropriate wavelength for consolidation will vary depending on the combination of particulate material and RAM used.

(60) Considering now FIG. **3***d*, this is a schematic diagram of the distribution sled **34** at four time-steps ta-td during the distribution sled forward stroke S**604**, where the pre-heater **24** is an array of pre-heaters. In FIG. **3***d* (*a*), the distribution sled **34** has moved across the build bed surface **12** such that the distribution device **37** has distributed a layer of particulate material **71** across the build bed

surface **12**. In this time-step, the pre-heater **24** has not yet passed over the location of the slice of the three dimensional object **2**. In FIGS. **3***d*(*b*)-(*d*), the pre-heater **24** located on the distribution sled **34** moves over the build bed surface **12** so as to irradiate and pre-heat the particulate material layer **71** in areas at and around the location of the slice of the three dimensional object **2**. It may be desirable to pre-heat an area greater than that of the slice of the object **2** (as shown), so as to prevent undesirable effects such as curl due to cooling of edge parts leading to in situ solidification of the consolidated material. In other implementations, depending on the properties of the particulate material, and the desired temperature profiles across the build bed or build bed surface **12**, the pre-heat step may be targeted to just the location of the three-dimensional object **2** in that particular slice of the object. In other implementations, it may be desirable to pre-heat all of the build bed surface **12**, but with differing intensities of irradiation so as to maintain a desired temperature or temperature gradient across the build bed surface **12**. This may be done, for example, to achieve improved properties of the three-dimensional object **2** by controlling the cooling rate after consolidation of a previous slice.

(61) In some implementations, the wavelength of the radiation emitted from the radiation sources **21** may be selected for pre-heating particulate material **71** on the build bed surface **12**. The desired pre-heat temperature depends on the particulate material being used, for example, a suitable pre-heat temperature range for PA12 is approximately 160-170° C.

(62) In some implementations, the radiation sources **21** in the radiation source assembly **100**/**101**/**102** may be chosen such that the wavelength of the radiation emitted by the radiation sources **21** is selected for both pre-heating and consolidating particulate material on the build bed surface **12**. The radiation sources may be light-emitting diodes (LEDs).

(63) In some implementations, the radiation source assembly **100**/**101**/**102** may comprise radiation sources **21** which are laser diodes.

(64) It may be understood that individual radiation sources **21** may be of a size that has been minimised so as to give fine granularity/per-pixel control of the region of the build bed surface **12** that they address when they are switched ON so as to consolidate and/or pre-heat particulate material **71** in a discrete region and enable formation of finely-featured objects **2**. To this end, the use of a focusing lens or lenses **23** enables even greater control of the region which is addressed by the radiation beams **26** emitted from the individual radiation sources **21**. The radiation source assembly may comprise one or more focusing lenses **23** which may comprise one or more cylindrical lenses. In other implementations, the radiation source assembly may comprise one or more focusing lenses **23** wherein the focusing lenses **23** may comprise one or more spherical lenses. The one or more focusing lenses may be arranged so each focusing lens is arranged to focus the beams **26** from a respective group of radiation sources **21**, or a sub-group thereof. The focusing lenses **23** may be used in implementations where it is desirable to direct the beams to specific points or discrete regions for example so as to increase the energy at the build bed surface.

(65) It may be understood that in some embodiments, as described with reference to FIG. **2***b* above, the apparatus **110** for manufacturing a three-dimensional object may comprise a first sled, distribution sled **34**, comprising a pre-heat source, and a second sled, printing sled **32** comprising a consolidation source, wherein the consolidation source and the pre-heat source are each provided by one or more of said radiation source assemblies **100**/**101**/**102**; or wherein the consolidation source is provided by said one or more radiation source assemblies **100**/**101**/**102** and the pre-heat source is provided by a (separate) pre-heater, such as pre-heater **24**. In other implementations, the apparatus **110** for manufacturing a three-dimensional object may instead comprise a sled comprising a pre-heat source and a consolidation source; wherein the consolidation source and the pre-heat source are each provided by one or more of said radiation source assemblies **100**/**101**/**102**; or wherein the consolidation source is provided by said one or more radiation source assemblies **100**/**101**/**102** and the pre-heat source is provided by a (separate) pre-heater, such as pre-heater **24**.

(66) The optional pre-heater **24** may then perform a pre-heat step to heat the particulate material

layer **71** to a constant temperature. The pre-heating may be controlled by the radiation system controller **53**, or the pre-heater may have its own separate controller. In some implementations, the pre-heater **24** may be a single heating source that spans the width of the build bed **16**, in other implementations, it may comprise an array of heat sources comprising one or more rows of said heat sources. The pre-heater **24** may, for example, be one or more ceramic lamps; halogen lamps; infrared lamps or light-emitting diodes (LEDs) or laser diodes. In some implementations, the pre-heater **24** may be a radiation source assembly comprising a plurality of radiation sources **21**(*i*-*vi*), each radiation source **21** being operable to emit a respective beam of radiation for pre-heating particulate material on a build bed surface **12** of the apparatus **110** for manufacturing a three-dimensional object; one or more collimators **22** arranged to collimate the beams **26** of radiation to produce a plurality of collimated beams **27**; and optionally one or more focusing lenses **23** arranged to focus the collimated beams **27** to create focused beams **28** that are focused onto particulate material **71** on the build bed surface **12**.
(67) In some implementations, the pre-heater **24** may be replaced by a fixed heater **29** located above the build bed surface **12** (see FIG. **2***a* and FIG. **7**) rather than being located on the distribution sled **34**. Said fixed heater **29** may be operated dynamically during the build process, for example by comprising an array of heaters which can be individually controlled ON/OFF to address desired portions of the build bed surface in accordance with image data and process steps. In some implementations, there may be both a fixed heater **29** and one or more pre-heaters **24** mounted on one or more sleds. In still other implementations, the pre-heater **24** may be omitted and a pre-heat step may instead be performed by the radiation sources **21** in the radiation source array **19** on the printing sled **32**. Said radiation sources **21** in the radiation source array **19** may be controllable to emit different levels of radiation such that they are able to emit lower levels of radiation in order to perform pre-heat steps as well as being able to emit levels of radiation sufficient to cause the particulate material to be consolidated. Such variability may be due to controllably emitting radiation at different intensities, or by being controllable to emit different types of radiation, e.g. radiation of different wavelengths, or by controlling the focusing lenses so as to alter the focal length of the beams and/or the spot size on the build bed surface.
(68) The radiation source assembly **100**/**101**/**102** may comprise array(s) **19** of radiation sources **21** wherein the radiation sources **21** are able to emit a plurality of wavelengths. For example in some implementations, the radiation source assemblies **100**/**101**/**102** may comprise different types of radiation sources **21** which are suited to different purposes, for example some that are suitable for pre-heating and some that are suitable for consolidation. The different types of radiation sources **21** may be arranged in separate groups and sub-groups in different parts of the radiation source array **19**, or they may be interleaved to form a multi-purpose radiation source array **19**. In other implementations, the different types of radiation sources **21** may be split between two or more radiation source arrays **19** which may be located at different places within the apparatus **110**.
(69) The radiation source assembly **100**/**101**/**102** as described above can be used in a method for manufacturing a three-dimensional object **2** from a particulate material **71**, whereby the method comprises depositing a radiation absorbing material (RAM) onto a layer of particulate material on a build bed surface, for example by moving one or more droplet deposition heads **33**, possibly arranged in an array **35**, across the build bed surface **12**; and then consolidating the particulate material **71** in the layer where the RAM has been deposited by moving a radiation source assembly across the build bed surface. In alternative implementations the RAM may be delivered to the build bed surface **12** by any suitable method, using any suitable apparatus. In still further implementations the RAM may be mixed with the particulate material **71** prior to its distribution across the build bed surface **12**; targeted heating in accordance with image data, using any of the embodiments and implementations described herein, then ensures that only the object **2** is consolidated.
(70) Attention is now drawn to FIGS. **4***a* and **4***b*, which depict a variety of ways in which a heater

25 (and optionally a pre-heater **54**) for a test apparatus for manufacturing a three-dimensional object can be operated in order to pre-heat or consolidate particulate material to form a slice of an object **2**. FIG. **4***a* depicts a pre-heater **54** and a heater **25** which can be moved across the build bed **16** in the X-direction in an arrangement similar to that widely used at present. Examples of such pre-heaters or heaters are one or more ceramic lamps; halogen lamps or infrared lamps. In this test example, which is not an embodiment, both the pre-heater **54** and the heater **25** span the width, in the Y-direction, of the build bed surface **12**. Both are ON for the entire traverse over the build bed surface **12** so as to pre-heat all the particulate material and to consolidate the object **2** which has been defined by depositing RAM in accordance with image data. As previously discussed, such an arrangement is used because current heaters **25**, and pre-heaters **54** take a considerable length of time to heat up to the desired temperatures, so cannot be effectively switched ON/OFF to heat only desired areas of the build bed surface **12**. It is clear that whilst the slice of the three dimensional object **2** where the RAM has been deposited may be consolidated using the heater **25**, the remainder of the particulate material **71** on the build bed surface **12** may receive undesirably excessive radiation which may lead to degradation of the particulate material, higher energy use and overall higher costs of production.

(71) FIG. **4***b* depicts a similar arrangement of a test apparatus to FIG. **4***a* except that both the pre-heater **54** and the heater **25** are OFF for the majority of the traverse in the X-direction, apart from when they are respectively passing across the pre-heat region **3** and the three dimensional object **2** on the build bed surface **12**. So the pre-heater **54** is OFF from X**0** to X**1**, ON from X**1** to X**4** and then OFF from X**4** to X**5**; whilst the heater **25** is OFF from X**0** to X**2**, ON from X**2** to X**3** and then OFF from X**3** to X**5**. It is clear that in the example given in FIG. **4***b*, less energy is supplied to the build bed surface **12** for both the pre-heat and the consolidation steps and the energy is more targeted in the X-direction at the region of interest. However, as for FIG. **4***a*, because both the pre-heater **54** and heater **25** span the build bed **16** in the Y-direction, there will be energy supplied to regions of the particulate material to either side of the three dimensional object **2** along the Y-direction, (between Y**0** and Y**1** and Y**4** and Y**5** for the pre-heater **54**, and between Y**0** and Y**2** and Y**3** and Y**5** for the heater **25**). Although only the area of the three dimensional object **2** has been defined by depositing RAM, and hence only this region will be consolidated, supplying energy to areas of the build bed surface **12** other than the location of the three dimensional object **2** and its surroundings for either pre-heat or consolidation may be undesirable as previously discussed.

(72) FIGS. **5***a* and **5***b* disclose embodiments aimed at reducing or preventing the above disadvantages of the test arrangements depicted in FIGS. **4***a* and **4***b*. Heater **25** has been replaced by a radiation source assembly **100**/**101** as depicted in FIG. **1***a* or **1***b*. Each of the individual radiation sources **21** in the radiation source array **19** can be controlled by a duty cycle such that they are switched ON when they pass above the slice of the three dimensional object **2**(*i*-*iii*) to be consolidated and which has previously been defined with RAM. A method for operating the radiation source assembly **100**/**101** may therefore involve addressing and controlling individual radiation sources **21** within the plurality of radiation sources **21**. For example such a method may involve one or more of controlling the intensity of the radiation being emitted and/or the wavelength(s) of the radiation being emitted and/or the spot size and/or the duty cycle of the plurality of radiation sources **21**(*i*-*vi*).

(73) For simplicity, the pre-heater **24** is omitted from FIGS. **5***a* and **5***b*, though it may be understood that it can be similarly arranged as an array of pre-heaters **24**(*i*-*n*), possibly as a radiation source assembly **100**/**101** as depicted in FIGS. **1***a* and **1***b* respectively, with the individual pre-heaters **24** switched ON or OFF so as to pre-heat regions of the particulate material layer **71** on the build bed surface **12** as desired. In other implementations, the radiation source assembly **100**/**101** may be used to both pre-heat and consolidate the particulate material layer **71** on the build bed surface **12** by controlling individual radiation sources **21** or groups and/or sub-groups thereof so as to provide different levels of radiation depending on whether they are performing a pre-heat

or consolidation step. This may be achieved by, for example, altering the radiation intensity, the power supplied to the radiation sources **21**, the focal length of the beam of radiation, the spot size on the build bed surface, the wavelength being emitted, etc., in response to control signals from the radiation system controller **53** and the overall process controller **60**.
(74) Turning now to FIG. **5***a*, the embodiment depicted comprises a multi-row radiation source array **19** where individual radiation sources **21** are controlled in a duty cycle such that they are switched ON when they overlie the part of the three-dimensional object **2** that they are addressing and OFF elsewhere. The arrangement **19**(*t*) of ON/OFF switching of the radiation sources **21** at three time instances t**1** to t**3** as the radiation source array **19** travels over the build bed surface **12** in the X-direction is shown.
(75) Considering now FIG. **5***b*, this depicts a different multi-row radiation source array **19** where the rows of radiation sources **21** are staggered. This may be advantageous in some implementations to provide a higher resolution for the areas of radiation of the build bed surface. The arrangement **19**(*t*) of ON/OFF radiation sources **21** at two time instances t**1** and t**2** in the duty cycle as the radiation source array **19** travels over the build bed surface **12** in the X-direction is shown. It may be understood that any suitable arrangement of staggered rows may be used, and that the embodiment depicted in FIG. **5***b* is merely a single non-limiting example. It may also be understood that for both the multi-row arrays **19** depicted in FIGS. **5***a* and **5***b*, the figures depict exemplary time instances and that in operation on a moving printing sled **32**, individual radiation sources **21** may be operated in a duty cycle such that they are ON as they pass over the location of the slice of the three-dimensional object(s) **2** on the build bed surface **12** that has been defined with the radiation absorbing material (RAM) and OFF once they have passed over it. It may further be understood that the pre-heater **24** may be an array of pre-heaters **24**(*i-n*) and may be arranged in either of the configurations depicted in FIGS. **5***a* and **5***b* and operated in a similar manner. In other implementations, it may be understood that the radiation sources **21** may be used for both a pre-heat step and a consolidation step, replacing or adding to the pre-heater(s) **24**. As such an embodiment of the present disclosure may comprise an apparatus **110** for manufacturing a three-dimensional object **2** from a particulate material, the apparatus comprising: a work surface **13** which comprises a build bed surface **12**; and one or more radiation source assemblies **100**/**101**/**102** as previously described, for pre-heating and/or consolidating particulate material **71** on the build bed surface **12**.
(76) FIG. **6** depicts an embodiment similar to that in FIG. **5***a* where the radiation source array **19** has been divided into two separate groups **55** and **56**. Group **55** comprises a row of radiation sources **21***a* that are suitable for pre-heating, distributed in the Y-direction so as to span the build bed **16**. Group **56** comprises two rows of radiation sources **21***b* that are suitable for consolidation, distributed in the Y-direction so as to span the build bed **16**. For example, the radiation sources **21***a* and **21***b* may be different types of LEDs. The arrangement **19**(*t*) of ON/OFF switching of the radiation sources **21***a*/**21***b* at three time instances t**1** to t**3** as the radiation source array **19** travels over the build bed surface **12** in the X-direction is shown. It can be seen that the pre-heat group **55** is controlled to irradiate a larger region than the slice of the three-dimensional object **2** (ON_**55**, dashed squares) and that the consolidation group **56** is controlled to address the slice of the three-dimensional object **2** that has been pre-defined with RAM (ON_**56**, solid squares). Within each group **55**, **56**, individual radiation sources **21***a*, **21***b* are controlled in a duty cycle such that they are switched ON when they overlie the part of the three-dimensional object **2** that they are addressing and OFF elsewhere.
(77) In alternative implementations, the two groups **55**, **56** may all comprise the same type of radiation source **21**, but the spot size or focal length of the focused beams **28** may be controlled for each group so as to alter the intensity of the focused beam **28** at the build bed surface **12**, or the first group **55** may be arranged with one or more collimators **22** to provide more diffuse heating at the build bed surface **12** and the second group **56** may be arranged with one or more collimators **22** and

one or more focusing lenses **23** to provide more focused heating of the build bed surface **12**. It may be understood that the arrangement of the two groups **55**, **56** depicted in FIG. **6** is in no way limiting and that in alternative implementations there may be a plurality of groups, and sub-groups, which may be arranged in any suitable manner and separately controlled by any suitable method, as desired. Further the individual radiation sources in the plurality of groups and/or sub-groups may not be arranged in discrete rows as depicted in FIG. **6**, but interleaved in a pattern, or arranged in alternate rows, or any other suitable layout.

(78) A method for operating the radiation source assembly **100**/**101**/**102** may therefore comprise arranging the radiation sources **21** into a plurality of individually addressable groups, each group comprising one or more radiation sources **21**. Further each group may comprise two or more radiation sources **21** and be divided into two or more sub-groups with each sub-group comprising one or more radiation sources **21**. The method may further comprise controlling a group or sub-group of the radiation sources **21** to pre-heat selected portions of the particulate material **71** on the build bed surface **12**. Additionally, the method may comprise controlling a group or sub-group of the radiation sources **21** to consolidate the particulate material on the build bed surface **12** that has been defined by deposition of a radiation absorbing material (RAM). The method may further involve the use of a radiation source assembly **21** that comprises one or more collimators **22** arranged to collimate the beams of radiation **26** emitted by the radiation sources **21** so as to produce a plurality of collimated beams **27**. The method may further involve the use of a radiation source assembly that comprises one or more focusing lenses **23**, and the method may further comprise controlling one or more focusing lenses **23** to focus the one or more collimated beams of radiation **27** onto the build bed surface **12**. The method may comprise controlling the radiation source assembly **100**/**101**/**102** so as to switch a group or sub-group of the radiation sources ON or OFF as they move over the build bed surface **12**. This may be in response to image data for the three-dimensional object **2**, or in response to temperature measurements of the build bed surface **12** performed by one or more temperature sensors wherein the temperature at discrete regions is higher or lower than a desired temperature for pre-heating or consolidating the particulate material **71**. The method of operation may further comprise controlling the radiation source assembly **100**/**101**/**102** so as to alter the amount of energy supplied to particulate material **71** at discrete regions on the build bed surface **12** by the radiation sources **21** in response to temperature measurements of the build bed surface **12** and information concerning the image data and desired temperatures.

(79) Turning now to FIG. **7**, this shows schematically an alternative arrangement for a pre-heat step where a fixed pre-heater array **29** located above the build bed surface **12**, such as that depicted in FIG. **2***a*, is used instead of or in addition to a movable pre-heater array **24** located on a sled. A method of manufacturing a three-dimensional object **2** may further comprise using a pre-heater to controllably pre-heat selected portions of the particulate material on the build bed surface **12**. Optionally, the pre-heater may be a pre-heater **24** such as radiation source assembly **100**/**101**/**102** as described herein and/or a fixed pre-heater array **29** located above the build bed surface. FIG. **7** depicts a build bed surface **12** and the ON/OFF arrangement of the pre-heater array **29**, which may be radiation source assemblies **100**/**101**, where each of the plurality of pre-heaters **29** can be individually addressed so as to controllably pre-heat required regions of the build bed surface **12** in accordance with the location of the slice of the object **2**. An array of collimating lenses and/or an array of focusing lenses as depicted in FIGS. **1***a* and **1***b* may be used with the fixed pre-heater array **29** so as to collimate and/or focus the beams of radiation produced by the pre-heaters **29**. The fixed pre-heaters **29** may comprise any suitable type of lamp or heater, and in some implementations may comprise radiation sources **21** as described herein, such as LEDs.

(80) It will be understood by those skilled in the art that most radiation sources emit a range of wavelengths rather than a single wavelength, and that the above disclosure encompasses radiation sources that emit a range of wavelengths, where the radiation source may be chosen to have a desirable wavelength range or spectrum or band in order to be suitable for pre-heating and/or

consolidating particulate material on a build bed surface. The term wavelength as used herein may therefore be understood to encompass radiation sources emitting a suitable range or spectra of wavelengths. References to a single wavelength may be understood to refer to a central wavelength within a distribution or range of wavelengths.
(81) Further it should be understood that spot size can be measured using a number of techniques such as setting up the optical arrangement and exposing a light sensitive film, or using a CCD (charge coupled device) or digital camera. This would be part of a set up calibration procedure whereby the spot size can be fine-tuned by, for example, moving the relative positions of the collimators **22** and/or the focusing lenses **23** and/or by focusing or defocusing them. This could be part of a set up calibration procedure before running an apparatus for the layer-by-layer formation of three-dimensional (3D) objects.
(82) It may be understood that any of the features described herein may be used in any combination and for any suitable purpose, with any of the other features described within the disclosure, as desired. For example a radiation source assembly **100** as per FIG. **1***a* may be used for a pre-heat step with a radiation source assembly as per FIG. **1***b* or FIG. **1***c* for a consolidation step.

Claims

1. An apparatus for manufacturing a three-dimensional object from a particulate material, the apparatus comprising: a piston configured to support a build bed of particulate material, the build bed having a build bed surface of particulate material; a distribution sled comprising a particulate material distribution device for distributing each layer of particulate material across the build bed; a printing sled comprising one or more droplet deposition heads operable to deposit a pattern of radiation-absorbing material (RAM) onto the layer, to define the cross-section of the three-dimensional object in said layer; a pre-heat source arranged to move across the build bed surface so as to preheat said layer, and a consolidation source arranged to move across the build bed surface so as to consolidate build material onto which RAM has been deposited while moving over the build bed surface; wherein the pre-heat source and the consolidation source each span a width, in a direction orthogonal to the direction of movement, of the build bed surface; and one or more radiation source assemblies for pre-heating and/or for consolidating particulate material on the build bed surface, the or each radiation source assembly comprising: a plurality of light-emitting diodes arranged in an array comprising a plurality of individually addressable groups, each light-emitting diode being operable to emit a respective beam of radiation towards the build bed surface; and one or more collimators, arranged to collimate the beams of radiation of the plurality of light-emitting diodes to produce one or more collimated beams of radiation, and to direct said collimated beams of radiation towards the particulate material on the build bed surface; wherein at least one of the pre-heat source and the consolidation source comprises one of said radiation source assemblies; and wherein the apparatus is configured to move the distribution sled while distributing each layer, the pre-heat source while pre-heating, the printing sled while depositing RAM, and the consolidation source while consolidating, in the same direction.
2. The apparatus according to claim 1, further comprising a fixed pre-heater for pre-heating particulate material on the build bed surface, wherein said consolidation source comprises one of said one or more radiation source assemblies, wherein said fixed pre-heater comprises a fixed pre-heater array located above the build bed surface.
3. The apparatus according to claim 1, further comprising: a first sled comprising the pre-heat source, the pre-heat source comprising a first radiation source assembly; and a second sled comprising the consolidation source, the consolidation source comprising a second radiation source assembly.
4. The apparatus according to claim 3, wherein said consolidation source comprises: one or more focusing lenses arranged to focus the beams of radiation onto the particulate material on the build

bed surface.
5. The apparatus according to claim 1, further comprising one or more focusing lenses arranged to focus the collimated beams of radiation onto the particulate material on the build bed surface, wherein said one or more collimators are arranged between the plurality of light-emitting diodes and the focusing lenses.
6. The apparatus according to claim 5, wherein the distance of the one or more focusing lenses from the build bed surface is adjustable.
7. The apparatus according to claim 1, wherein the distance of the light-emitting diodes from the build bed surface is adjustable; and/or the distance of the one or more collimators from the build bed surface is adjustable.
8. The apparatus according to claim 1, wherein the plurality of light-emitting diodes comprises: a first group of light-emitting diodes operable to emit radiation within an absorption band of wavelengths of the particulate material so as to preheat, and a second group of light-emitting diodes operable to emit radiation within an absorption band of wavelengths of the RAM so as to consolidate the particulate material to which the RAM has been provided.
9. The apparatus according to claim 1, further comprising one or more temperature sensors configured to measure the temperature of the build bed surface, wherein the one or more radiation source assemblies are configured to be operable in response to temperature measurements by the one or more temperature sensors.
10. A method for manufacturing a three-dimensional object from a particulate material in an apparatus for manufacturing the three-dimensional object from particulate material, the apparatus comprising a build bed surface of particulate material, and one or more radiation source assemblies for pre-heating and/or for consolidating particulate material on the build bed surface, the or each radiation source assembly comprising: a plurality of light-emitting diodes arranged in an array spanning the width of the build bed surface and comprising a plurality of individually addressable groups, each light-emitting diode being operable to emit a respective beam of radiation towards the build bed surface; and one or more collimators, arranged to collimate the beams of radiation of the plurality of light-emitting diodes to produce one or more collimated beams of radiation, and to direct said collimated beams of radiation towards the particulate material on the build bed surface; wherein the method comprises at least one of the steps of: moving the radiation source assembly across the build bed surface while operating one or more of the plurality of light-emitting diodes so as to preheat the particulate material; and/or depositing a radiation absorbent material (RAM) onto a layer of particulate material on the build bed surface by moving one or more droplet deposition heads across the build bed surface; and/or consolidating the particulate material in the layer where the RAM has been deposited by moving the or a further radiation source assembly across the build bed surface while operating one or more of the plurality of light-emitting diodes.
11. The method according to claim 10, further comprising individually addressing and controlling the plurality of light-emitting diodes while moving them across the build bed surface.
12. The method according to claim 10, further comprising controlling one or more of the intensity of the radiation being emitted and/or the wavelength of the radiation being emitted and/or the duty cycle of said plurality of light-emitting diodes.
13. The method according to claim 10, wherein the one or more radiation source assemblies further comprises one or more focusing lenses arranged to focus the collimated beams of radiation onto the particulate material on the build bed surface, and wherein said one or more collimators are arranged between the plurality of light-emitting diodes and the focusing lenses, the method further comprising adjusting the focal length(s) and/or the spot size of said one or more focusing lenses.
14. The method according to claim 10, wherein the plurality of light-emitting diodes of the radiation source assembly are arranged into a plurality of individually addressable groups, each group comprising one or more of the light-emitting diodes, the method further comprising moving the radiation source assembly across the build bed surface while operating a group of the one or

more groups of light-emitting diodes to carry out the step of pre-heating by pre-heating selected portions of the particulate material on the build bed surface.
15. The method according to claim 14, comprising the step of depositing the RAM onto the layer of particulate material on the build bed surface by moving the one or more droplet deposition heads across the build bed surface, and consolidating the particulate material in the layer where the RAM has been deposited, the method further comprising moving the radiation source assembly across the build bed surface while operating a further group of the one or more groups of the light-emitting diodes to carry out the step of consolidating the particulate material on the build bed surface that has been defined by deposition of the RAM.
16. The method according to claim 14, further comprising controlling whether a group or sub-group of the light-emitting diodes is switched ON or OFF as they are moved over the build bed surface.
17. The method according to claim 10, wherein the apparatus further comprises one or more temperature sensors arranged to monitor the temperature of the build bed surface, the method further comprising controlling the radiation source assembly so as to alter the energy supplied to selected portions of the particulate material on the build bed surface in response to temperature measurements performed by the one or more temperature sensors.
18. A method for manufacturing a three-dimensional object from a particulate material in an apparatus for manufacturing the three-dimensional object from particulate material, the apparatus comprising a build bed surface of particulate material, and a fixed pre-heater array located above the build bed surface, the fixed pre-heater array comprising one or more radiation source assemblies for pre-heating particulate material on the build bed surface, the or each radiation source assembly comprising: a plurality of light-emitting diodes comprising a plurality of individually addressable groups arranged in an array, each light-emitting diode being operable to emit a respective beam of radiation towards the build bed surface; and one or more collimators, arranged to collimate the beams of radiation of the plurality of light-emitting diodes to produce one or more collimated beams of radiation, and to direct said collimated beams of radiation towards the particulate material on the build bed surface; wherein the method comprises controllably pre-heating one or more selected portions of the particulate material on the build bed surface using the fixed pre-heater array.